

Supporting Information

θ -Fe₃C dominated Fe@C core-shell catalysts for Fischer-Tropsch synthesis: roles of θ -Fe₃C and carbon shell

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Experimental results.

Characterization of MIL-101(Fe)

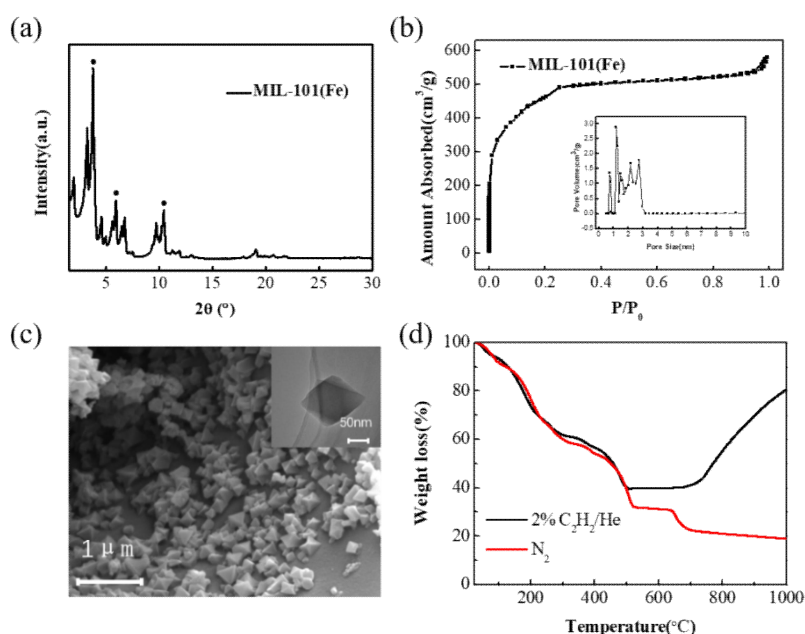


Fig. S1. Characterization of MIL-101(Fe). (a) XRD pattern (b) N_2 physisorption isotherm and pore size distribution (insert) (c) SEM image and TEM image (insert) (d) TG curves under 2% C_2H_2/He and N_2

MIL-101(Fe) is constructed from $Fe_3(\mu_3-O)(NO_3)(H_2O)_3(BDC)_3$ clusters as secondary building units and the dicarboxylates as linear linkages, with an infinite structure with three dimensional connected channels. In this research, instead of commonly used hydrofluoric acid, nitric acid was used as mineralizer to synthesis MIL-101(Fe).

Fig. S1 shows the characterization results of MIL-101(Fe). The crystallite structure of MIL-101(Fe) was examined by XRD, as shown in Fig. S1a. The diffraction line in Fig. S1a corresponds well to the characteristics of MIL-101($2\theta=3.8^\circ$, 5.9° , 10.4°), confirming the successful growth of MIL-101(Fe).[1] The N_2 physisorption isotherm and pore size distribution of sample are shown in Fig. S1b. The surface area of MIL-101(Fe) is as high as $1632 m^2/g$. Fig. S1b displays a type-I isotherm typically

associated with micropore. The average pore size is 1.9 nm. The morphology of MIL-101(Fe) is examined by SEM and TEM, shown in Fig. S1c. The SEM image and TEM image (insert) show well-defined octahedron morphology with an average size of about 150 nm.

A TG analysis was carried out to study the pyrolysis behavior of MIL-101(Fe). Fig. S1d shows the TG curves of MIL-101(Fe) heated in N_2 and 2% C_2H_2/He , respectively. The two curves exhibit two stages of weight loss. The initial weight loss corresponds to the loss of physically adsorbed solvents (water and DMF), occurred between room temperature and 300 °C. The second mass loss is assigned to decomposition of linker at 300-500 °C. Then a platform emerges at 500-700 °C in C_2H_2 and 500-650 °C in N_2 . The weight loss in C_2H_2 is 10% less than that in N_2 . It is caused by the significant decomposition of ethyne at 500 °C. A clear difference is observed in the regions above 700 °C. A weight loss occurs in N_2 caused by phase transformation while a weight augment in C_2H_2 corresponding to excessive carbon deposition.

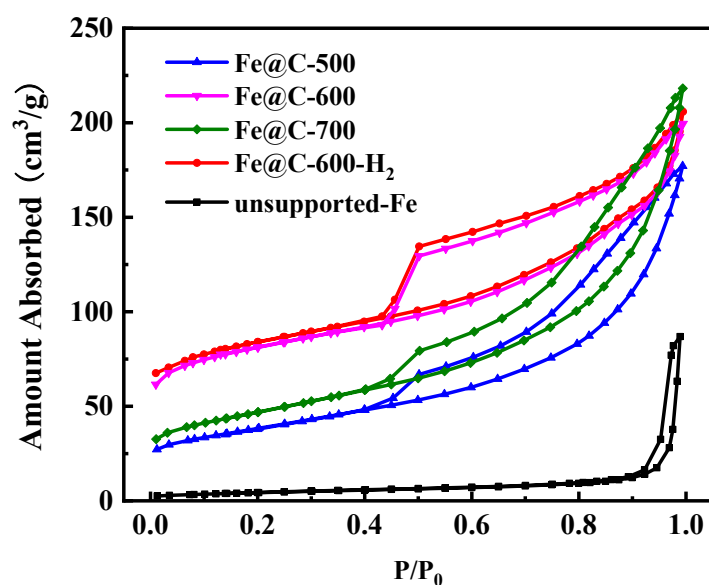


Fig. S2. N_2 physisorption isotherm of the catalysts

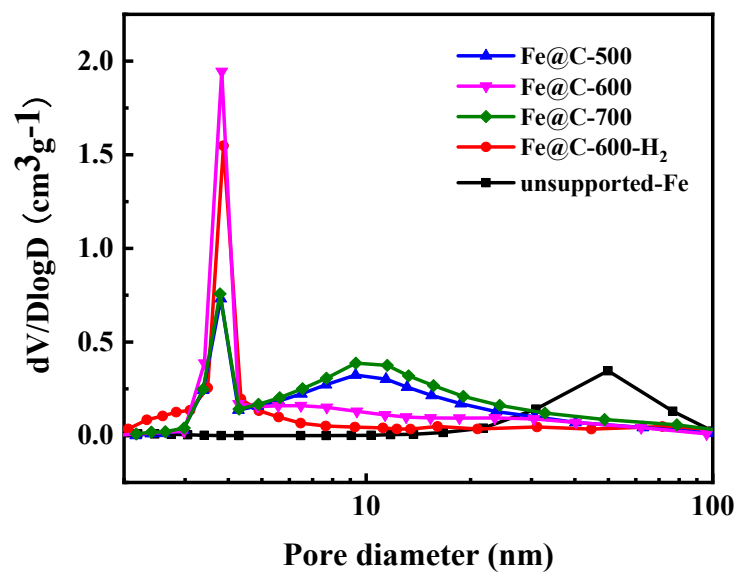


Fig. S3. Pore size distribution of the catalysts

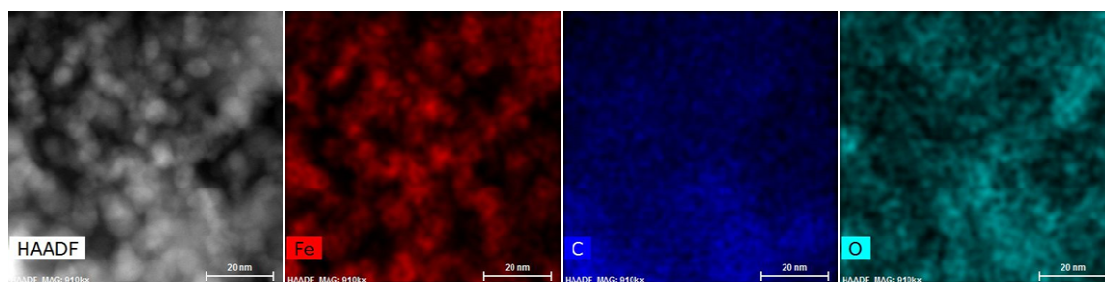


Fig. S4. Elemental mapping of Fe@C-500

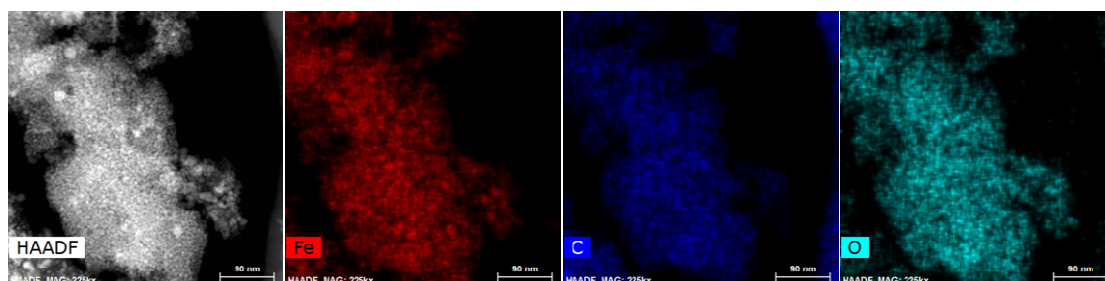


Fig. S5. Elemental mapping of Fe@C-600

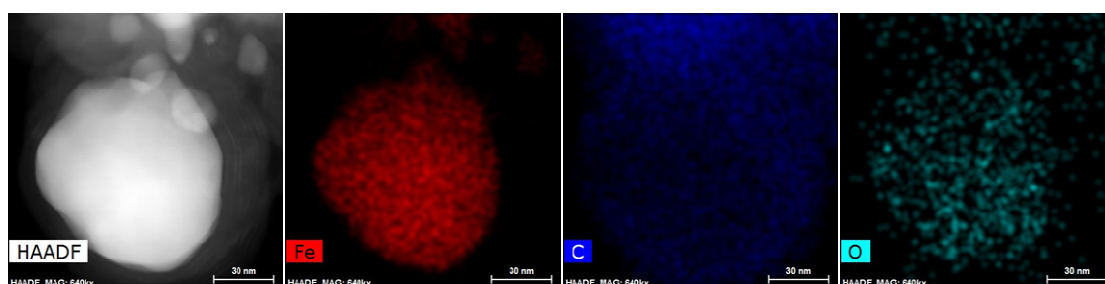


Fig. S6. Elemental mapping of Fe@C-700

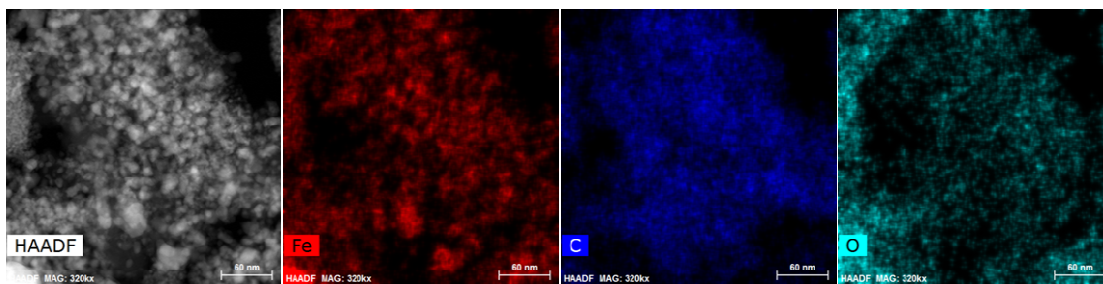


Fig. S7. Elemental mapping of Fe@C-600-H₂

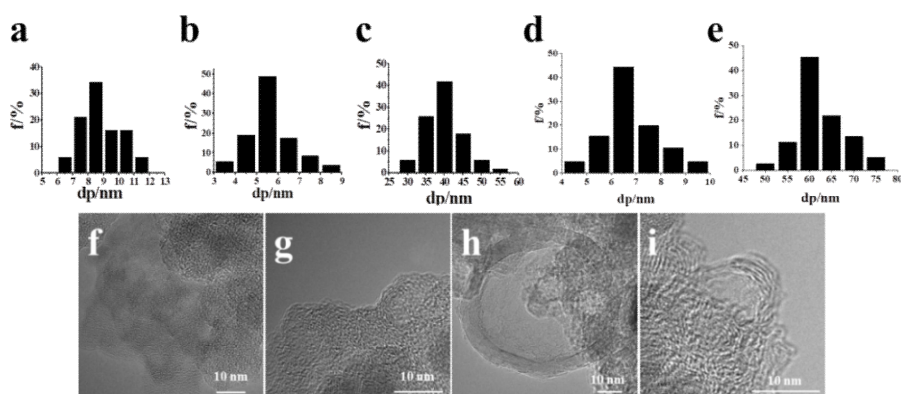


Fig. S8. Particle size distributions (a-e) and HRTEM images of carbon residues after acid picking for three days of catalysts (f-j). Fe@C-500: (a) (f); Fe@C-600: (b) (g); Fe@C-700: (c) (h); Fe@C-600-H₂: (d) (i); unsupported-Fe: (e).

Table S1. Fitting parameters of the Raman spectra

Catalyst	peak	Center max	FWHM	Max height	I _D /I _G
Fe@C-500	1	1346.7	64.3	316	1.10
	2	1584.8	64.3	288	
	3	2698.6	84.3	178	
Fe@C-600	1	1209.1	93.5	311.7	0.79
	2	1333.6	144.5	1501.7	
	3	1515.1	160.5	312.1	
	4	1600.6	45.5	1896.3	
	5	2565.8	175.3	110.9	
	6	2690.1	145.3	139.3	
	7	2900.3	265.3	276.8	
Fe@C-700	8	3188.4	72.0	80.7	0.91
	1	1236.3	125.3	157.1	
	2	1338.0	95.3	626.2	
	3	1488.2	160.3	151.2	

	4	1597.6	61.3	686.1	
	5	2698.5	126.9	167.3	
	6	2920.6	180.5	95.2	
Fe@C-600-H ₂	1	1222.4	176.3	515.7	0.78
	2	1338.4	131.4	1247.6	
	3	1505.6	205.1	596.3	
	4	1600.9	56.6	1599.6	
	5	2620.1	270.6	530.5	
	6	2693.6	71.7	389.2	
	7	2904.5	226.2	577.1	
	8	3176.9	111.7	444.3	

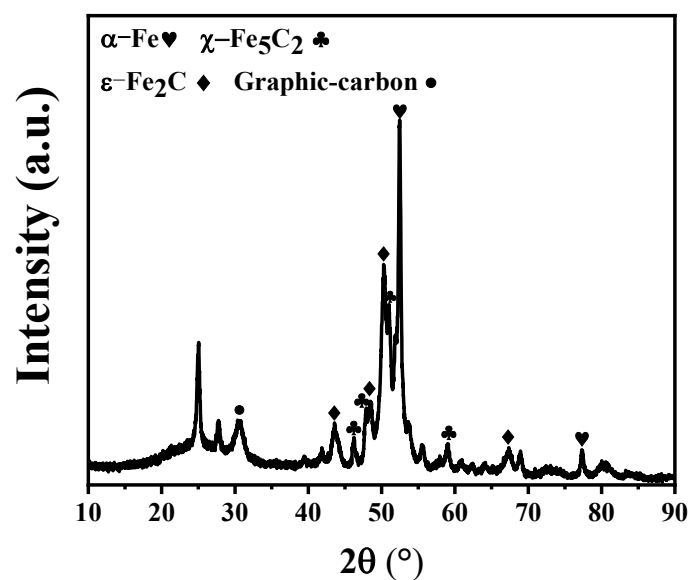


Fig. S9. XRD patterns of the Fe@C-600-H₂ after FTS reaction

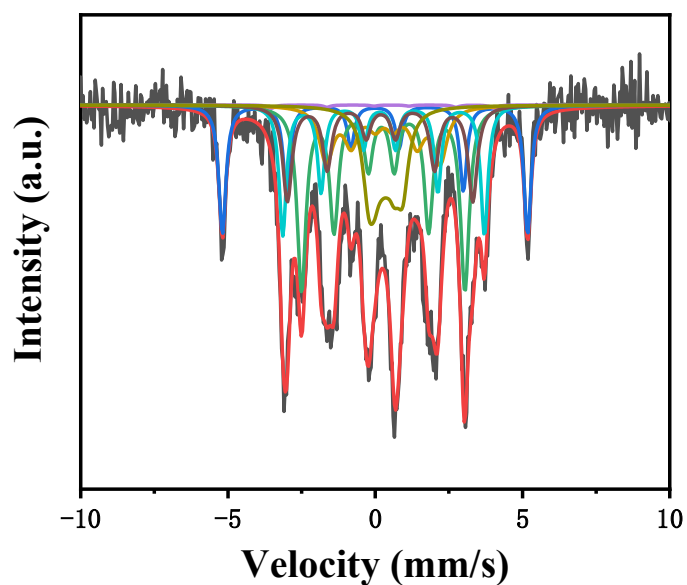


Fig. S10. Mössbauer spectra of Fe@C-600-H₂ after FTS reaction
 Table S2. Mössbauer parameters of Fe@C-600-H₂ after FTS reaction

Catalysts	IS (mm/s)	QS (mm/s)	Hhf (kOe)	Phase	Area(%)
Fe@C-600-H ₂ after FTS	0.21	0.14	213	χ -Fe ₅ C ₂ (I)	15.7
	0.18	-0.03	196	χ -Fe ₅ C ₂ (II)	14.6
	0.33	0.33	160	χ -Fe ₅ C ₂ (III)	13.8
	0.24	0.07	172	ε -Fe ₂ C(I)	14.2
	0.4	-0.3	235	ε -Fe ₂ C(II)	13.0
	0.29	-0.03	120	ε -Fe ₂ C(III)	13.2
	-0.01	-	327	α -Fe	15.5

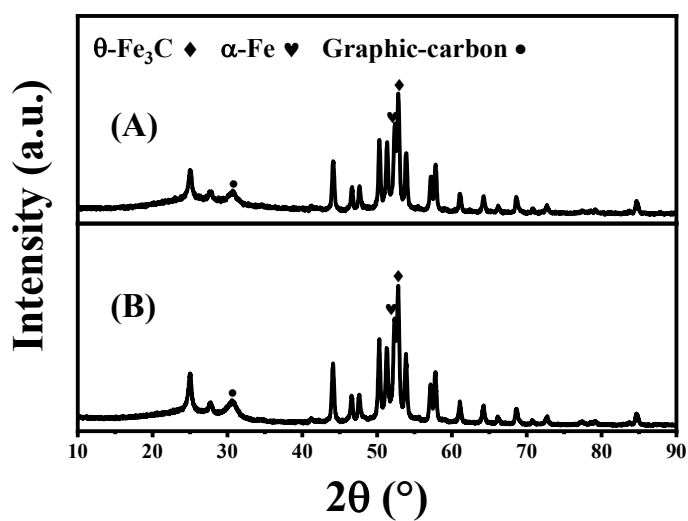


Fig. S11. XRD patterns of the Fe@C-600 after FTS reaction for (A) 50 h (B) 120 h

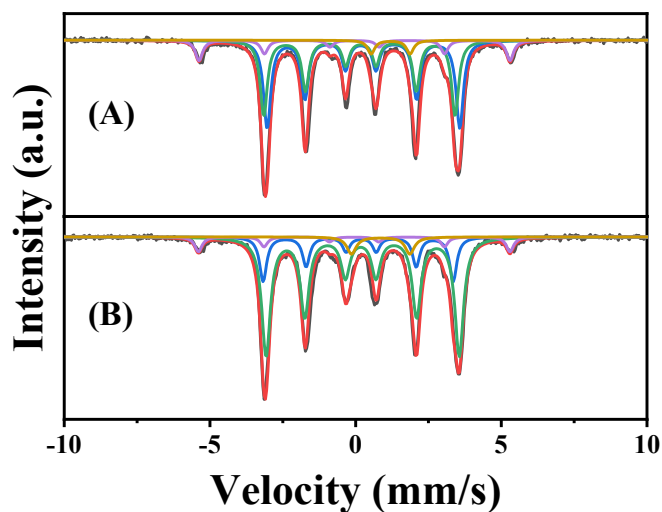


Fig. S12. Mössbauer spectra of Fe@C-600 after FTS reaction for (A) 50 h (B) 120 h

Table S3. Mössbauer parameters of Fe@C-600 after FTS reaction

Catalysts	IS (mm/s)	QS (mm/s)	Hhf (kOe)	Phase	Area(%)
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Fe@C-600 after 50 h FTS	0.22	0.09	208	θ -Fe ₃ C(I)	46.9
	0.15	-0.04	206	θ -Fe ₃ C(II)	41.4
	0	-1.43	—	spm Fe ⁰	2.0
	1.2	-1.33	-	Fe ²⁺	3.6
	-0.03	-	331	α -Fe	6.1
Fe@C-600 after 120 h FTS	0.13	-0.1	208	θ -Fe ₃ C(I)	47.3
	0.21	0.06	206	θ -Fe ₃ C(II)	41.6
	0	-1.66	—	spm Fe ⁰	2.3
	0.86	-2.00	-	Fe ²⁺	3.1
	-0.05	-	332	α -Fe	5.7